SDU
University of Southern Denmark

An actor-network theory instrument for design practitioners

Ağça, Ayşe Özge ; Buur, Jacob

Published in:
The International Journal of Designed Objects

DOI:
10.18848/2325-1379/CGP/v17i02/49-64

Publication date:
2023

Document version:
Accepted manuscript

Document license:
CC BY

Citation for pulished version (APA):
Ağça, A. Ö., & Buur, J. (2023). An actor-network theory instrument for design practitioners. *The International Journal of Designed Objects*, *17*(2), 49-64. https://doi.org/10.18848/2325-1379/CGP/v17i02/49-64

Go to publication entry in University of Southern Denmark's Research Portal

Terms of use
This work is brought to you by the University of Southern Denmark.
Unless otherwise specified it has been shared according to the terms for self-archiving.
If no other license is stated, these terms apply:

- You may download this work for personal use only.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying this open access version

If you believe that this document breaches copyright please contact us providing details and we will investigate your claim.
Please direct all enquiries to puresupport@bib.sdu.dk

Download date: 06. Nov. 2026

An Actor-Network Theory Instrument for Design Practitioners

Ayşe Özge Ağça, University of Southern Denmark, Denmark
Jacob Buur, University of Southern Denmark, Denmark

Abstract: In this paper, we explore a novel idea of theory instruments: tangible artifacts that encourage participatory design practitioners in the industry to reflect on interactions with technological devices in the light of a particular theory. Our Actor-Network Rings instrument is based on Actor-Network Theory as this is one way to challenge taken-for-granted assumptions about what effects digitalization have on product users and their lives. The instrument consists of wooden rings, wooden balls, clothespins, and magnets. By adding wooden clothespins as actors and plastic ones as actants on wooden network rings, upholding a wooden ball becomes visible and even tangible as a program of action. Shifting or removing clothespins visualizes concepts of imaginary substitution and delegation. We analyze two cases of how designers and design researchers in companies use the instrument to investigate dilemmas of people and technology. With transcripts from video documentation and hand-drawn visualizations, we show that the instrument elicits broader insights into the dilemmas and even opens ideas for future scenarios. We argue that there is a potential for developing such theory instruments that, in a playful manner, offer fundamental theory perspectives to inform understandings of digitalization.

Keywords: *Digitalization, Tangible Tools, Participatory Design, Future Scenarios*

Introduction

With increasing digitalization, design researchers and designers in the industry are struggling to take into account the effects that interactive products have on users and their lives. In a study of UX practices in manufacturing companies, Kjærsgaard et al. (2021) argue that this is not simply a problem of finding better methods; rather, it is a more fundamental question of theoretical perspectives. To help practitioners gain broader perspectives, we have started experimenting with theory instruments—tangible objects that work as conversation starters to challenge the taken-for-granted assumptions about people and technologies. In this article, we analyze how one of these instruments encourages practitioners to understand the situation of their users or their own in a new light. We call the instrument "Actor-Network Rings" (Figure 1) as it is based on Actor-Network Theory (ANT).

Actor-Network Theory

ANT developed in response to the proliferation of electric appliances everywhere post WWII. As anthropological research focused on the systems and their hierarchical control mechanisms (Dant 2005), a distinct theory emerged to explain the complexity of these objects and systems relations. ANT is particularly associated with Bruno Latour, who claimed that the social influence of machines should be accounted for (Latour 1992). He

asks, “When we act, who else is acting? How many agents are also present? How come I never do what I want? Why are we all held by forces that are not of our own making?” (Latour 2005, 43). Latour describes how action is a knot under the invisible influence of something, but when it slowly unravels, it becomes a combination of many agencies. While these agencies carry out their actions in a complex and heterogeneous manner, this ‘overtaken’ situation occurs unconsciously in taken-for-granted assumptions. Actions should be kept as “a surprise, a mediation, an event” (Latour 2005, 45). Callon (2001) states that ANT is an approach that emerged in the 1970s within the sociology of scientific knowledge, aiming to explain the process of how scientific facts and technical artifacts are produced, as well as how their validity and efficacy are established and understood. With this aspect, ANT may help researchers to gain deeper insight about this invisible and unconscious effect of artifacts and their surprise actions.

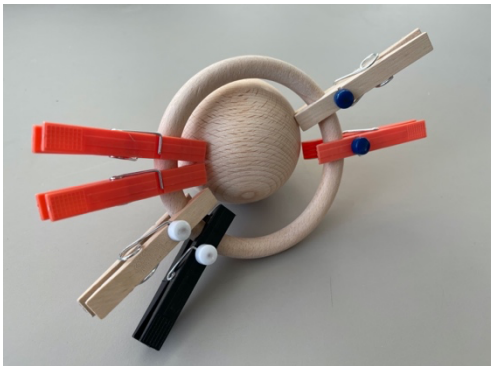


Figure 1: Actor-Network Rings Theory Instrument.
Source: Buur et al. 2022

ANT challenges us to look at the non-human worlds and their relations with the human worlds. We can figure out how to talk about complexity, appreciate complexity, and practice complexity (Mol 1999, 82; Law 1999, 10). Dant (2005, 76-81) suggests that material things have some measure of agency that gives the power to do or act toward or in the company with people. According to Gell (1998, 16), an agent is one “who causes events to happen in their vicinity.” *The* agency is associated with individuals and objects that are perceived as initiating specific types of causal sequences, meaning events that result from deliberate actions or acts of the mind. The agency is what the artifact does through its user, maker, owner, or material. Hence, in Actor-Network Theory, actions are “borrowed, distributed, suggested, influenced, dominated, betrayed, translated” by agents such as by the *actors* who are pushed to act in a certain way by the *actants* (Latour 2005, 46). Callon (2001) explains this network as active and collective in human and nonhuman worlds. Including nonhumans in the collective, ANT has the potential to inspire researchers about the roles of technology and science in their construction and evolution. Pinch and Bjiker's study (1984) leads to understanding not just the flexibility of how people interpret objects but also the

flexibility of how these objects are designed. This makes the theory highly appropriate also to discussing relations between digital products and people. Observing the connections and interactions established among the agents of an active network, as well as how they collectively change their roles (distribution of competencies, prescription), will assist in generating alternative scenarios (delegation, imaginary substitution) in response to these potential changes.

The Participatory Design Tradition

Our theory instruments take inspiration from prior work on tangible tools in the Participatory Design tradition. Brandt (2006, 181–182) suggests the term "things-to-think with" for tangible artifacts in participatory workshops. Mitchell and Buur (2010) find several benefits of playing with tools in their study of tangible business models: tangible tools can facilitate systemic thinking, produce simplicity, convey the dynamism of the business, make it simpler to think large, inspire new connections and associations, help narrative telling, and make it simple to recall experiences. Buur and Gudiksen (2012) show how tangible tools encourage business practitioners to hypothesize and think in alternative (business) scenarios. Kobbelgaard, Bødker and Kanstrup (2020) identify design games as investigating challenges and possibilities, addressing early design specifications, and mapping potential design spaces. They see design games as ideal instruments for analyzing existing and future actions, scenarios, or practices, with notable literature insights in this field. Design games come in a wide variety, but in general they tend to make field material (e.g., user research) available for participants to make sense of through manipulation and clustering—but with little emphasis on theoretical grounding. Our theory instruments almost do the opposite: They bring a theory perspective on field material, but do not directly embed field experiences—that is for the participants to bring in.

Jansen (2014) and Dragicevic, Jansen, and Moere (2019) represent data in physical form to provide new perspectives. To Jansen (2014, XXV), "instruments are a specific way of looking at controls through the lens of the instrumental interaction model." The term instrument suits us well, as it denotes something playful and explorative, in contrast to the tool, which signals efficient production. This is how we prefer to see theory work—as an explorative trying-out of different perspectives. Where Participatory Design has generally focused on empowering "users" in engaging with design activities, our experiments with theory instruments scaffold (design) practitioners in making sense of user activities (Case 1) or of their own practices (Case 2).

The questions we want to answer with our experiments are:

> (1) Can a tangible instrument support practitioners in (re)thinking about how they understand the fundamental relations between people and (interactive) objects?
> (2) Can we think about this as a challenge of how (digital) products are constructed in connection to individuals on personal and organizational levels?

The Design Process of Actor-Network Rings

We want to explain how we developed this theoretical instrument and the process of giving physical form to the theory. We were contacted by a healthcare company with a request to provide support in analyzing their user research data (Buur et al. 2023). While they had been through Post-it clustering activities, they were not convinced of their explanatory power and requested a stronger theoretical foundation. In the first session, we introduced a set of theory cards (Figure 2, left) explaining each theory's core. We then discussed deeper using Post-it notes and visuals (middle). The concept of cards was well-received for challenging preconceived notions; however, the participants necessitated significant facilitation, and they did not encourage as much palpable interaction as we had hoped for.

While Latour (1992) says we should "compare machines with texts" to understand choices of roles and actions between human and nonhuman worlds, we noticed that we need to build in made-up actants and actors and distribute the skills of tangible tools for a theory between the selected material. Taking inspiration from the simple graphic on the card (Figure 2, right), we wondered how we might weave physical objects together to form a network. We needed an object that could hold on to things, grab them, carry them, and form a web or structure in different shapes, materials, or colors. Clothespins have the potential to provide both physical and action diversity in a semantic sense. We, therefore, explored different rings, spheres, and other shapes.

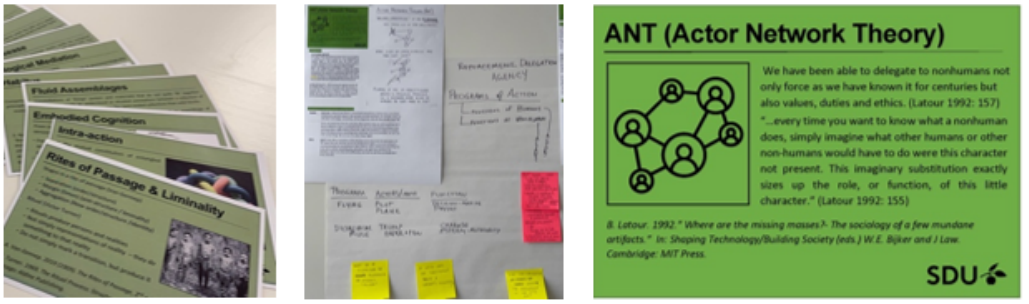


Figure 2: Theory Cards (Left); Actor Network Theory Discussions (Middle); Actor Network Theory Card (Right)
Source: Kjærsgaard et al. 2021

The theory instrument we developed would challenge participants to easily distinguish between actors and actants (colored clothespins) while bringing them together in the same network (ring). Hopefully, the simple handling would encourage participants to enact different stories of people with technological objects in an explorative, experimental mode (Figure 3). We also enclosed wooden spheres to help establish programs of action.

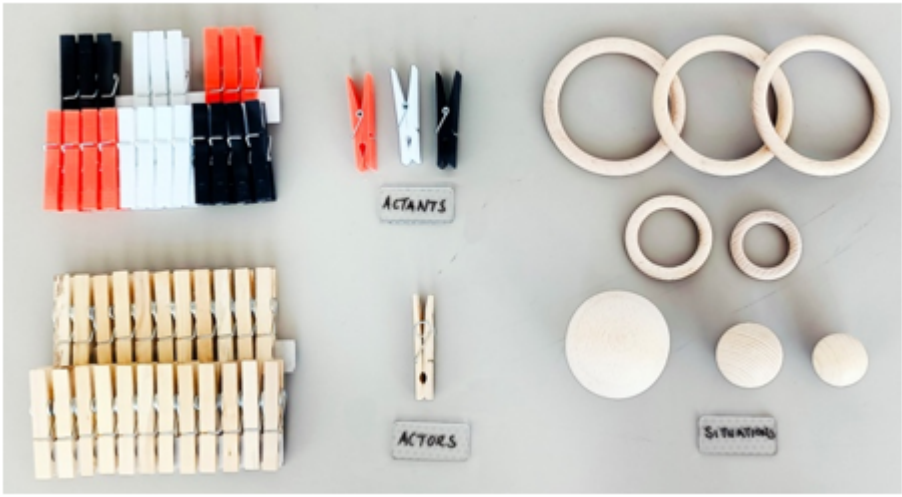


Figure 3: The Theory Instrument Parts and Their Meanings
Source: Ağça

Methods

We trialed the theory instrument in workshops with industrialists from several manufacturing companies, with master and doctoral students from different universities, and with visiting researchers, and we observed vivid engagement throughout (see Appendix 1). The workshops were documented with video recordings (with participants' permission), photos, and notes. For this article, we selected two cases that illustrate the process well: a half-day workshop with design researchers at a conference and a two-hour workshop in a robotic systems company. Facilitation was guided with written instructions (as in Buur et al. 2023) to achieve some consistency. Each workshop had three participants, who were encouraged to choose a people-technology dilemma that could be relevant to investigate. In both cases, we asked the participants to first select clothespins to represent relevant actors and actants, and then to place them together around a wooden ring. One or more wooden spheres were then suggested to represent a program of action. During the process, facilitators would prompt the participants to play with imaginary substitution and delegation or prescription.

For the analysis, we prepared two kinds of transcripts: a verbal transcription of the conversations in the video and a graphical transcription in the form of action drawings of each construct—what the participants said and what they made. Since this is a very new form of instrument, we are still in the early stages of the evaluation process.

Following each session, we also invited the participants to evaluate the activity and the theory instrument in a focus group interview. For some of the sessions, we additionally asked participants by email to give their opinions after a week.

Case 1: E-Car Charging

In the design researcher workshop, we provided field material on electric car charging: a video summary of observations, a textual field note from a study, and an autoethnographic account. Having studied the material, we challenged the participants (P1, P2, P3) to express with the ANT instrument how people and things were involved in the charging process. One participant started by distinguishing different environments of charging with wooden balls and wooden rings around them (Figure 4, left).

The big ball is the main house where you spend more time, the medium size is the summer house (that figured in the field material), and the small size is away-from-home chargers. She attached two wooden clothespins as actors: the user and her child. After establishing this main structure, she invited the other two participants to update the story of the electric car user.

In the main house, they chose colored clothespins as a car, cable, and charging box and wrote names or symbols on the pins to remember them. Then the participants moved the actors and actants to the vacation house, almost like playing a doll-house game:

> P1: When they need to go to their vacation house, mom and child [takes one wooden clothespin from the big ring]
> P2: With their car. [shows the car with his fingers].
> P1: So, they have the cable that also goes with her. Now she knows how to take it away, so it charges. So the phone and the clever box stay at home. [indicates cutting the connection by separating between clothespins such as the main house phone and electrical company in the vacation house]. They are now all on vacation, but they have to go charge the car eventually. So they have to drive to try it for free. Now she knows how to do the cable, but these boxes are not everywhere that they want to go. Right.
> P2: So this now charges.
> P1: It's no longer frustration. They have now learned that this is okay.

While adding clothespins, they maintained the relations between each place by interlinking the pins. When one network broke, they noticed that this turned emotional for the actors or created problems. When they changed the delegated roles of actants, it helped to reestablish the network again (Figure 4, right).

> P1: Yeah, then it's less happiness here. And it's still related to the amount of charge, depending on the distance they have to go. There is still a discussion. Can we now charge 100% because we are going to drive this separate relationship destination? Now, there are theories that become important, so let's put the theories as a concept [add white clothespins with theory name tag].

> P2: How long does it take to charge the surcharges that we have now at three points? Now, you should be able to charge that.

The participants introduced the concept of *theories* as an *actant*:

> P1: [T]he theories around charging if it should be full charge or 80%. And this was what, temperature, the time of the year when they go to these different locations because the winter is heavier and uses the battery more. Thus, it's more important to understand what distances you can get from there.

The theories that car owners develop are about how much they need to charge the car and when. In a sense, the participants came to discuss new responsibilities of the actants:

> P2: So you can really change these actors and their importance depending on the perspective you take and tell a new story with the same actors, actants, and networks. Same but different every time, right?

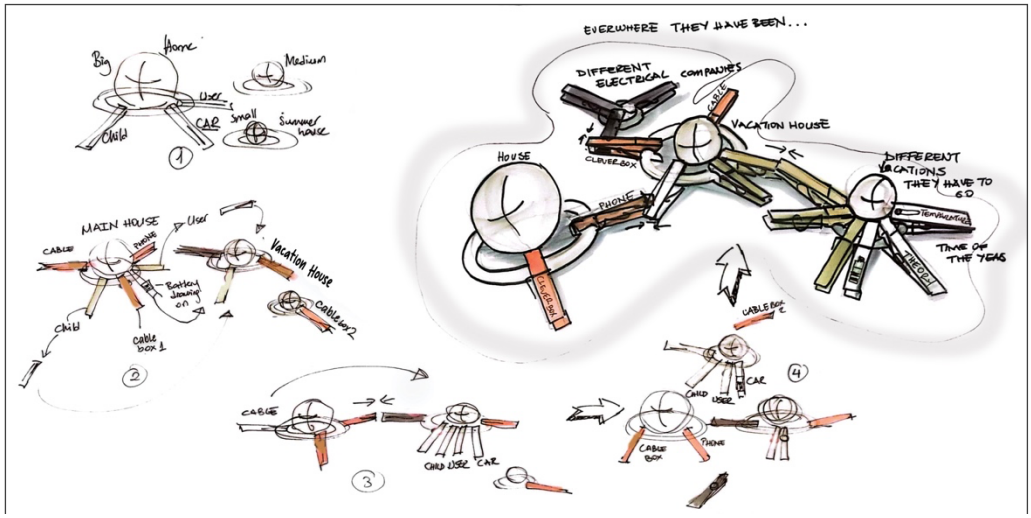


Figure 4: How Digital Products Are Constructed (Left); How Multiple Networks Are Relating Each Other (Right)
Source: Agca and Buur

Throughout the twenty-minute activity, the participants developed a rather nuanced perspective on how people and technologies interact in the charging processes. They became rather confident discussing technology as actants that exhibit agency in an ANT sense. They did perhaps not use the ANT terminology fully, but we can see from the transcripts that they talked about prescription, delegation, and imaginary substitution. The concept of networks is likely less refined than in ANT because of the simple materials (pins are either gripping a network ring or not). Still, the general theory perception seems to fuel and support the investigation well.

Case 2: Online Strategy Meeting

In the workshop with the robotic systems company, the participants chose to investigate what happened at a recent, conflictual strategy meeting where, according to them, both technologies (online) and documentation formats played a role. The participants are a business strategist (P1), a design engineer (P2), and a UX designer (P3). The company develops professional service robots for health care, construction, and agriculture. The theme of the meeting was the development of a new technological feature. In the beginning, the participants added wooden clothespins (actors) to a large ring to recap who was there (Figure 5, left).

> P1: What happened was the Autonomy Department came in [adds a wooden pin].
> P2: The Autonomy Department started asking you what this is about (…) because they need to work on this feature, but they don t know what it is about.
> P1: Yes, exactly. Because the feature is way down the road, but (…) looking into the future, they would like to know what they should be working on, so they know how to plan their work and do their programming and all that (…) They came in and said: "What is this thing?" [points to pin].
> P2: Can we define it? What the Autonomy Department always asks is "Why is it important? And what is the problem we are solving?"
> P3: You are also here as a Strategy Department (adds another wooden pin] and UX Department [P1 adds another pin because P2 was also involved in this meeting]
> P1: Then the Product Management [adds a pin].
> P2: They are two persons – they are connected [added a second manager pin].
> P3: But this person was not there [touches pin] (…) He is the CTO who is standing here [detached pin, set it upright next to ring] [laughter].

This introduction helps us to understand a bit of the organizational structure. It aligns well with how Latour uses ANT to explain how technological developments and the distribution of facts and artifacts happen through negotiation between various groups (Fallan 2010). Such groups can be seen as actors who construct different interpretations, agendas, needs, and desires as networks through enrolling and persuading. Cresswell, Worth, and Sheikh (2010, *6*) explain ANT as how and why things occur by exploring relationships. We see that the new feature decision in the robotics company is not important enough for the CTO to engage, even if it runs across a number of departments. Apparently, there isn't agreement about whether the feature is really valuable enough:

> P2: One of the product managers said, "Actually, it would be nice, but it has a very small weight" [pulls in a smaller wooden ring and pushes it back again].

> P1: As a UX person, I would ask the people here [points to managers], "Why do you need this smart function?" I would start with this question because I can see everybody is lost in the conversation of "Why do you think we need that?"
> P2: There was one more person in that meeting. The Technical Lead of the project [adds another pin]. He is the person responsible for the core functionality. So, he asked actually the resources to make sure it happened. And we are all supplying to him [points from the other pins to him].
> P1: But he is kind of "Yeah, I can do whatever" [shrugs his arms in an I-don't-care gesture].
> P2: "You just have to figure it out!" [laughter].

We sense that the UX person felt a responsibility to pick up the discussion about why the feature should be developed, but it is also tied into a power play between managers. So far, the participants have traced the actors in the meeting. When challenged by the facilitator to consider the role of actants (e.g. technologies), the participants started discussing the constraints of the online setup but quickly moved on to the role of documentation as more important (Figure 5, right). P1 unclipped all pins, cleaned all parts of the tool, and they started to explain the process of documentation with actant pins:

> P1: I got the notes from P2. THAT's my problem now [he slams the ring with his own pin into the table].
> P2: [Adds an orange actant pin) So one of the tools we used was documentation. We were documenting at least some of the discussion.
> P1: But we also have another one [added a white pin] that's the product roadmap that started the whole thing …because the feature was on there.
> P2: [Replaces white with an orange pin] Because it's documentation, they are the same.
> P2: And this is part of that one [points to roadmap pin and documentation pin]. It's actually two things [adds another orange pin) This is the roadmap, this is the description of what we want to solve, and this is actually the second slide, where we document what this is about [points to each pin in turn]. This is the summary, and this is the internal note. And this one [removes the internal note pin] will be forgotten when we have solved it – when this is gone, it is unnecessary. This is actually the description of the problem, which is then a detail in this main document [points in turn to pins].

This was a very vivid moment in the session. The participants, in turn, tried to illustrate what role the various documents as actants play in their interactions at the meeting and afterward. They then bring in the (heavy) wooden ball:

P1: It could be that now the upper management comes in [drops the wooden ball on the table] "New strategy, guys – that is not a problem anymore"
P2: Yes, or new customer [spins a ball in a ring] Customer says – maybe it's not the paying customer, a smaller customer – says, "Yeah, I really want this" [indicating a different ring].

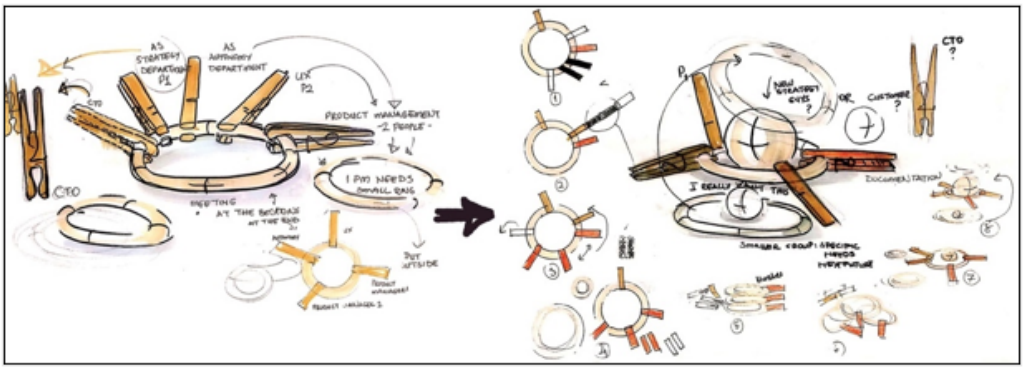


Figure 5. The ANT Rings Show Relationships between Professionals and Documentation (Left) with Steps of Ideation for All Changes with the Help of Interaction Systems (Right)
Source: Agca and Buur

During the process, the participants engagedly imitated the voices of other employees in a story-telling manner; they seemed to try the perspective of the others and shape the ring accordingly. We see this as a powerful example of playing the instrument. The participants used the material to recall their own experiences, reflect on them in light of ANT actor and actant relations, and project into the future to see new possibilities for the delegation of roles between actors. The more they role-played, the more they noticed details in the role of documentation. Latour (1992) explains delegation occurs between actants and actors or actants and actants. For Yaneva (2009), delegating action to non-humans means that humans substitute artifacts for acts of people to shape their program of action by redistributing competencies and prescribing responsibilities. The behavior urged back to actors by delegation is named prescription (Latour 1992).

Discussion

How Does the Theory Instrument Engage Participants?

Clothespins

In both cases, the participants easily accepted that the wooden clothespins represented actors and the plastic ones actants (non-humans). The diversity and number of nonhumans were surprisingly big in both cases. The instrument triggered discussions about what can be actants: in the first case, theories about charging, and in the second, documentation. Both stood out as becoming important to understand what happens. While the participants did

not use the ANT-term imaginary substitution, there were incidents where participants took away an actor or actant pin and discussed what this could mean to the situation. When we examine the table in the Appendix, it becomes evident that wooden clothespins, as represented in individual experiences such as personal energy consumption, health issues, or e-car charging, predominantly symbolize family members and friends. However, in discussions with companies, they tend to represent company personnel and relationships among them. Similarly, plastic clothespins have been associated with more personal items and office-related products or systems.

Rings

In the first case, the rings were sometimes described as space, sometimes as departments, and sometimes as layers of progression. In the second case, the participant used rings to show who was there. In order to show more than one network and the current conditions, all available rings of different sizes were used. In other cases (see Appendix 1), when examining personal experiences, the rings have represented the home, its spaces, or other visited locations. During discussions with companies, on the other hand, they embody the agency of various elements such as time, space, intra-company relationships, or objects. Participants opted for rings of varying sizes based on the significance or dimensional variations of the locations they defined.

Wooden Spheres

The spheres can be manipulated, such as rolling, and be placed in the rings. While here they had less of an active role, we have seen cases where participants more clearly used the pins to uphold a sphere as a program of action in ANT terms. This required a good deal of balancing and triggered conversations about what "upholding" actually means.

What Does Playing with the Instrument Trigger?

In both cases, the participants co-created the models by recapping either information they learned from field observations or experiences they shared from a recent meeting. The instrument seems to encourage the participants to identify with the actors' roles and enact possible future scenarios. We asked the participants to reflect on the experience. One of the design researchers from Case 1 points out how this instrument pushes them to play with the theory:

> P1: It worked very well where things happened at different places, and these places were connected (…) all these things we could sort of incorporate into building and rebuilding and reconfiguring the system (…) that we could discuss, and we could model.

Another Case 1 participant reflects on how the objects are open for interpretation:

> P3: We've seen different temporalities, different categories of objects that interact (…) [points to the networks they created]. See, it's open, (…) this is a technology that allows us to do things, but also to think with things. [Picked up a sphere] The instruments are like open stuff and like meaningless stuff. (…) So, thinking with the objects allows us to weave materiality into what we are discussing.

In a follow-up mail, a robotic company practitioner praised their workshop as an Actor-Network Game. And even when meeting one year later, they still clearly remembered the session with the instruments "These tools are actually metaphors (…) your way to talk about new things and to frame our thinking by touching something."

How Does the Instrument Encourage Practitioners?

Through the analysis of these sessions, we discerned that the instrument inspires practitioners to see in a new light the situation of product users or their own organization. In Case 1, we understand that more people are involved, such as spouses, neighbors, and customer service. Moreover, the participant thinks that people develop their own theories of charging:

> P2: And these are random locations where they have to go, and there are issues with how much to charge the battery and the theories around charging if it should be full charge or 80%.

In Case 2, this new light is the moment that the participants realize the importance of documentation in negotiations:

> [Management leaves]
> P2: So I started writing, writing, writing because I wanted something for these people [points to manager pins] to return to. (…)
> P1: I got the notes from P2. THAT's my problem now [he slams the ring with his own pin into the table].
> P2: [Adds an orange actant pin] So one of the tools we used was documentation. We were documenting at least some of the discussion. (…)
> P1: But we do also have another one [adds a white pin] that's the product roadmap that started the whole thing …because the feature was on there.

Another interesting urge the instrument prompts to the online strategy meeting is challenging taken-for-granted assumptions about people and technologies. During the meeting, the management department looked like they knew what they wanted from the beginning but still let the other departments do brainstorming. In the end, the management makes decisions for them.

> P1: And they [product management] were saying [holds the two manager pins] "Hmmm, the way it's kind of scoped today, maybe it's not adding that much value anymore". So that was kind of – funny. To some extent.
> P2: As a UX person, I would ask the people here [points to managers], "Why do you need this smart function?" I would start with this question because I can see everybody is lost in the conversation of "Why do you think we need that?".

Based on all these challenges and enlightenments, the last thing we observed was that the instrument helps practitioners gain broader perspectives. For instance, the e-car charging happens in many locations, not just at home, and the participant explains by grabbing different sizes of balls in her hand:

> P1: We also could put different things in the center. And these balls then represented either the home where you spend most of the time and then versus summer house where you spend some time versus a small location where you start very little time.

Does the instrument manage to convey genuine Actor-Network Theory? The theory instrument is a gross simplification of ANT. It conveys very basic concepts and relations and appears to do this well. The finer details of ANT's relational ontology are beyond the limitations of what the instrument can show. The instrument does not constitute a full representation of ANT; it is a conversation starter that encourages a particular perspective on people and technology.

Conclusion

We have shown how a tangible theory instrument can encourage practitioners in a playful manner to adopt a different theoretical perspective on the role of people and technology. The Actor-Network Rings is one of several ideas for theory instruments in the making. We hope that we will also be able to show how such instruments can encourage beneficial shifts between several theoretical perspectives.

This is an analog instrument. It will only prove limited value in scaffolding online meetings, as participants tend to be less flexible in procedure and mutual improvisation. In physical meetings, even if we suggested specific definitions of what the instrument's components represent and supplied guidelines on how they should be played, participants would often improvise applications or reverse steps while touching the materials.

The experiments with Actor-Network Rings show that this instrument does not just start the conversation but also helps practitioners gain broader perspectives, encourages them to see their own or general situations in a new light, and challenges taken-for-granted assumptions about technological dilemmas of people.

Acknowledgment

We would like to thank our company partners and workshop participants for willingly exploring the potential of our theory instruments and the graduate students in the IT Product Design 2021 program for running some of the company workshops. The project was supported by It-vest Networking Universities in Denmark.

Informed Consent

The authors have obtained informed consent from all participants.

Conflict of Interest

The authors declare that there is no conflict of interest.

REFERENCES

Brandt, Eva. 2006. "Designing Exploratory Design Games: A Framework for Participation in Participatory Design." *Proceedings of the Ninth Conference on Participatory Design: Expanding Boundaries in Design* 1: 57–66. https://doi.org/10.1145/1147261.1147271.

Buur, Jacob, and Sune Gudiksen. 2012. "Interactive Pinball Business." In *Proceedings of the 7th Nordic Conference on Human-Computer Interaction: Making Sense Through Design*, 129–38. https://doi.org/10.1145/2399016.2399036.

Buur, Jacob, Mary Karyda, Mette Gislev Kjærsgaard, Jessica Sorenson, Ayşe Özge Ağça, and Michela Antonelli. 2023. "A Collection of Tangible Theory Instruments for Design Anthropology." In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction*, 1–12. https://doi.org/10.1145/3569009.3572799.

Callon, Michael. 2001. "Actor Network Theory." In *International Encyclopedia of the Social & Behavioral Sciences* 62–66. Amsterdam: Elsevier. https://doi.org/10.1016/b0-08-043076-7/03168-5.

Cresswell, Kathrin M, Allison Worth, and Aziz Sheikh. 2010."Actor-Network Theory and Its Role in Understanding the Implementation of Information Technology Developments in Healthcare." *BMC Medical Informatics and Decision Making* 10 (67). https://doi.org/10.1186/1472-6947-10-67.

Dant, Tim. 2005. *Materiality And Society*. Maidenhead: Open University Press.

Dragicevic, Pierre, Yvonne Jansen, and Andrew Vande Moere. 2021. "Data Physicalization." In *Handbook of Human Computer Interaction*, 1–51. https://doi.org/10.1007/978-3-319-27648-9_94-1.

Fallan, Kjetil. 2010. *Design History: Understanding Theory and Method*. London: Berg.

Gell, Alfred. 1998. *Art and Agency: An Anthropological Theory.* Oxford: Clarendon Press.

Jansen, Yvonne. 2014. “Physical and Tangible Information Visualization.” Doctoral diss., Informatique: Université Paris-Sud.

Kjærsgaard, Mette Gislev, Wafa Said Mosleh, Jacob Buur, and Jessica Sorenson. 2021. “Anticipating Connectivity in (UX) Design Practices: Reframing Challenges by Introducing Theory Cards.” In *Ethnographic Praxis in Industry Conference Proceedings* 1: 159–173. https://doi.org/10.1111/epic.12065.

Kobbelgaard, Frederik Victor, Susanne Bødker, and Anne Marie Kanstrup. 2020.“Designing a Game to Explore Human Artefact Ecologies for Assistive Robotics.” In *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society* Article 27: 1–10. https://doi.org/10.1145/3419249.3420181.

Latour, Bruno. 2005. *Reassembling the Social: An Introduction to Actor-Network-Theory*. Oxford: Oxford University Press.

Latour, Bruno. 1992. "Where Are the Missing Masses? The Sociology of a Few Mundane Artifacts." In *Shaping Technology / Building Society: Studies in Sociotechnical Change*, edited by Wiebe E. Bijker and John Law, 225–258. Cambridge, MA: MIT Press.

Law, John. 1999. “After Ant: Complexity, Naming and Topology.” *Sociological Review* 47 (1): 1–14. https://doi.org/10.1111/j.1467-954x.1999.tb03479.x.

Mitchell, Robb, and Jacob Buur. 2010. “Tangible Business Model Sketches to Support Participatory Innovation.” In *Proceedings of the 1st DESIRE Network Conference on Creativity and Innovation in Design*: 29–33.

Mol, Annemarie. 1999. “Ontological Politics. A Word and Some Questions.” *Sociological Review* 47 (1): 74–89. https://doi.org/10.1111/j.1467-954x.1999.tb03483.x.

Pinch, Trevor J., and Wiebe E. Bijker. 1984. “The Social Construction of Facts and Artefacts: Or How the Sociology of Science and the Sociology of Technology Might Benefit Each Other.” *Social Studies of Science* 14 (3): 399–441. https://doi.org/10.1177/030631284014003004.

Yaneva, Albena. 2009. “Making the Social Hold: Towards an Actor-Network Theory of Design.” *Design and Culture* 1 (3): 273–288. https://doi.org/10.1080/17547075.2009.11643291.

ABOUT THE AUTHORS

Ayşe Özge Ağça: Ph.D., Department of Design, Media and Educational Science, University of Southern Denmark, Kolding, Denmark
Corresponding Author’s Email: ayoz@sdu.dk

Jacob Buur: Professor, Department of Design, Media and Educational Science, University of Southern Denmark, Kolding, Denmark
Email: buur@sdu.dk

Appendix A

Workshops with Actor-Network Rings Instrument

Case	*Participants*	*Actors*	*Actants*	*Program of Action*
Healthcare Company	1 researcher 1 student 1 industrialist	patient, nurse, doctor, family, pharmacists, distributors	insulin pens, tracking device, documents, internet, alcohol wipes, calendar, fridge, soda	Manage diabetes
Healthcare Company	4 industrialists	patient, spouse, restaurant assistant	car, cruise control, healthy snacks, drive-through, fast food	Manage diabetes diet
Water Systems Company	3 industrialists	installer, technicians, interns, staff, team leaders, manager, water utilities researcher, users, non-users	water meter, transformer, omni filter, computer, program, tools, plumping station	Read water meter
Robotics Company	3 industrialists	CTO, strategists, UX designer, managers,	documentation, computer, internet, product broad map	Set strategy in online meeting
Design Consultancy	3 consultants	speaker, audiences, intern, employees	techno devices, dishwasher, papers, chairs	Facilitate online meeting
Citizen Science Project	3 masters' students	participants, families, friends, colleagues, teachers	phones, computer, applications, social media, electronic devices	Track personal energy consumption
Phd Course	3 PhD students	participants, family, parents	home, travels, bathroom, kitchen, shower, hot water, watch, alarm, toothbrush, coffee& toast machine	Reduce private energy consumption
Postgrad Course	3 high-school teachers	driver, support technician	cable, charger, support system,	Facilitate e-car charging
Drs 2022 Workshop	3 researchers	user, children, electrical companies	house, summer house, phone, temperature & charge apps, cable, cable box	Manage e-car charging